

Reso: Enhancing Social Communication for the Deaf & Hard-of-Hearing

MOIZZA AZHAR, Université Paris-Saclay, France

PEIWEN ZHANG, Université Paris-Saclay, France

BILL TANG, Université Paris-Saclay, France

DANIYAL NASIRI-BAVIL, Université Paris-Saclay, France

OSUKE SASHIDA, Université Paris-Saclay, France

Online meetings have become essential for collaboration, yet they remain largely inaccessible for Deaf and Hard-of-Hearing (DHH) individuals. This paper introduces Reso, a visualization system designed to enhance social communication by making emotional cues visible.

Additional Key Words and Phrases: Accessibility, DHH, Online Meetings, Captions, Emotion Visualization

1 Introduction

Online meetings have created new accessibility barriers for Deaf and Hard-of-Hearing (DHH) users, especially because standard captions only convey spoken words and not emotion, tone, or energy [4, 10]. This limits social presence and makes it harder to follow the intent behind a speaker’s message, particularly in fast-paced group conversations [6, 11].

Current accessibility tools often focus on transcription or speaker tracking, but they rarely represent the emotional and rhythmic qualities of speech [7, 16]. As a result, DHH users may still experience cognitive overload and caption fatigue when trying to follow multiple visual cues at once [12, 13].

To address this gap, we introduce **Reso**, a videoconferencing interface that visualizes emotion and vocal intensity through color-coded captions and sound-rhythm cues [9, 15]. Our goal is to make online communication more expressive, socially meaningful, and accessible for DHH users [2, 3].

2 Related Work

The shift toward video-mediated work has created new accessibility barriers for Deaf and Hard-of-Hearing (DHH) users, particularly in professional collaboration and online meetings. Prior studies show that DHH participants often experience cognitive overload when relying on visual-only interfaces, especially when captions, speaker cues, facial expressions, and other visual elements compete for attention at the same time [8, 14]. To reduce these barriers, researchers have recommended clearer teleconferencing practices such as speaker name announcements, interpreter pauses, and structured turn-taking, while also recognizing that DHH groups often develop their own social accessibility norms and captioning etiquette in practice [11–13].

Captioning remains one of the most common accessibility solutions, but standard text captions often fail to support communication in dynamic or small-group settings. Prior work has explored meeting visualizations such as word clouds and mobile captioning through head-mounted displays to help DHH users manage attention and reduce visual dispersion, although these solutions can introduce distraction or device-related constraints [1, 6, 7]. More recent studies

Authors’ Contact Information: Moizza Azhar, moizza.moizza@universite-paris-saclay.fr, Université Paris-Saclay, Orsay, France; Peiwen Zhang, peiwen.zhang@universite-paris-saclay.fr, Université Paris-Saclay, Orsay, France; Bill Tang, bill.tang@universite-paris-saclay.fr, Université Paris-Saclay, Orsay, France; Daniyal Nasiri-Bavil, daniyal.nasiri-bavil@studenti-paris-saclay.fr, Université Paris-Saclay, Orsay, France; Osuke Sashida, Sashida.osuke@universite-paris-saclay.fr, Université Paris-Saclay, Orsay, France.

suggest that accessibility support should be adapted to the communication context rather than relying on a single generic captioning approach [10].

Another major limitation of current systems is their inability to convey emotional and paralinguistic cues. Plain captions typically preserve the words of speech but lose the tone, emphasis, and affect that shape how messages are understood socially. To address this gap, researchers have proposed richer captioning systems that visualize prosody, emotion, and emphasis through changes in font, motion, size, or color, with studies showing that these approaches can improve the understanding of speaker intent [4, 9]. Multimodal approaches have further extended this direction by adding haptic feedback, which can enhance speaker identification and emotional engagement in real time [3, 15].

This work also aligns with a broader shift toward Deaf-centered design, where accessibility technologies are shaped by Deaf culture, identity, and lived experience rather than by medicalized assumptions alone [2, 14]. In this view, the goal is not only to translate speech into text, but to design communication tools that preserve social presence, emotional nuance, and meaningful participation in collaborative settings [5, 8].

3 System Design

3.1 Design Goals

The system was designed as an accessibility-oriented captioning interface for Deaf and Hard-of-Hearing users during online spoken interaction. Its primary goal was to extend conventional captions beyond verbal transcription by adding visual cues for affective tone and conversational activity. In this way, the prototype aimed to address both the difficulty of inferring emotion from plain text and the cognitive effort required to monitor rapid conversation.

A second goal was to preserve readability while adding expressive information. The interface therefore used a small set of consistent visual mappings rather than a dense display of speech features. A third goal was to support iterative prototyping and evaluation, allowing the system to be refined through repeated testing of caption generation, emotion assignment, and overlay presentation.

3.2 Interface Components

The prototype was organized around two complementary components: a control interface and a transparent caption overlay. The control interface supported caption input, system monitoring, and display adjustment, while the overlay remained visually unobtrusive during communication.

The overlay served as the main presentation layer, keeping captions visible while other meeting applications were in use. Captions were presented as discrete visual units so that utterances could be read quickly, and a rhythm-oriented display was included to provide a lightweight indication of conversational activity. Together, these components created a layered representation of communication in which textual meaning, emotional tone, and interactional pacing could be perceived in parallel.

3.3 Visual Encoding of Emotion and Prosody

The central design strategy of the prototype was to encode emotional interpretation through color. Each caption was assigned to one of six categories: happy, sad, angry, surprised, neutral, or fearful. These categories were associated with distinct visual treatments so that users could identify likely tone at a glance. Warm and vivid colors were used for more activated states, while cooler or more subdued tones were used for calmer states.

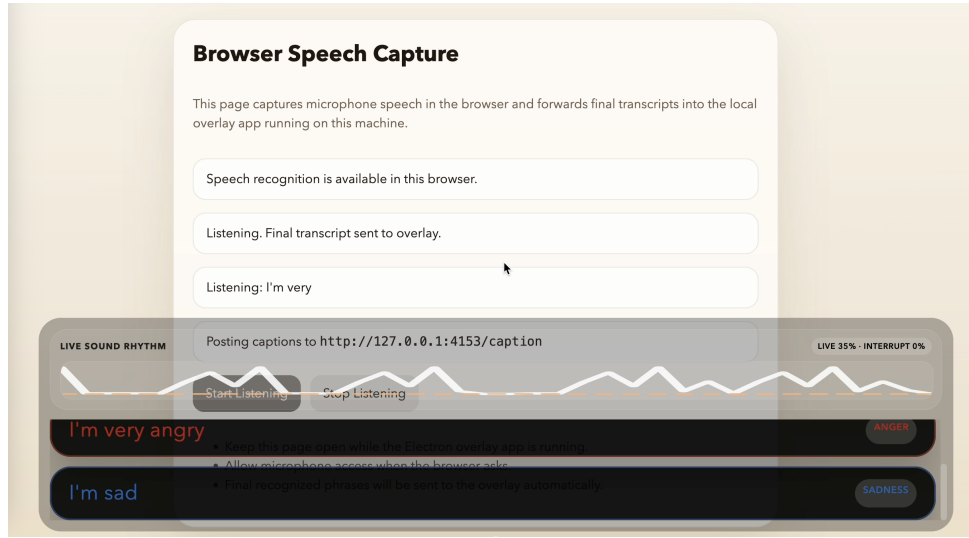


Fig. 1. Prototype user interface showing the control panel and the transparent emotion-coded caption overlay.

This encoding was intended as an interpretive aid rather than a definitive statement of speaker emotion. The category set was intentionally constrained to provide accessible contextual information while maintaining visual consistency.

In addition to color-coded emotion labels, the interface included a visual rhythm layer representing conversational activity in simplified form. Rather than presenting detailed acoustic measurements, this layer conveyed general changes in speaking intensity or pacing. In the present prototype, this rhythm visualization should therefore be understood as a supportive awareness cue that complements the emotion-coded captions.

3.4 Prototype Implementation

The prototype was implemented as a desktop-based caption overlay system for real-time online communication. At a high level, the system followed a sequential flow in which spoken or simulated utterances were represented as caption text, passed through an emotion-tagging stage, and then rendered in the transparent overlay. This architecture allowed the study to focus on how emotional and prosodic information could be communicated visually through captions.

Emotion assignment in the current version was primarily text-driven. Linguistic cues within each caption were used to estimate affective orientation and map the result into the reduced set of interface categories. This choice supported stable prototyping and allowed the visual design to be evaluated independently of a fully developed multimodal recognition pipeline.

The prototype should therefore be understood as a proof-of-concept system that operationalizes the proposed design ideas in an interactive form. More advanced audio recognition, prosodic feature extraction, and multimodal fusion remain directions for future development.

4 Methodology

We conducted a within-subjects online user study to compare Reso with a standard captioning baseline, using short pre-recorded videos in which a speaker delivered a single sentence.

4.1 Participants

We recruited 18 participants aged 18 to 30. All were hearing individuals without reported auditory impairments. Because Reso is intended for Deaf and Hard-of-Hearing (DHH) users, hearing participants were used as proxy users to evaluate usability, visual clarity, and cognitive load.

4.2 Experimental Conditions

The study was hosted on a custom web-based platform deployed through GitHub Pages. Participants accessed the experiment using their personal computers or smartphones in their own environments. The stimuli consisted of short pre-recorded video clips, each containing a single speaker delivering one sentence. Each clip was paired with one of the two interface conditions, allowing direct comparison between the baseline and the proposed multimodal design.

The experimental interface was designed to be lightweight and browser-based so that it could be accessed without any specialized installation. This choice was made to reflect realistic remote-use conditions and to reduce barriers to participation.

The study used a within-subjects design with one independent variable: Interface Type. Each participant completed both conditions in counterbalanced order to reduce ordering effects.

- **Condition A: Standard Captions.** This baseline condition displayed plain white text captions below the video. No color coding, rhythm visualization, or additional visual cues were provided.
- **Condition B: Emotion-Coded Captions + Sound Rhythm.** This experimental condition displayed captions with emotion-based colors and a continuously updated frequency graph representing voice intensity. Participants first selected an emotion, then received visual feedback through the colored caption and rhythm display before confirming their final answer.

Figure 2 illustrates both interfaces side by side, showing the difference between the baseline and the multimodal design.

4.3 Procedure

The study began with participant ID entry, informed consent, and a short instruction page explaining the task. Participants then completed two blocks of six videos each, with condition order randomly counterbalanced across participants. Videos were presented continuously with no interval between trials. The task procedure was identical across both conditions: after each video finished playing, participants made their response using the interface for that condition.

4.4 Measures

We collected three quantitative measures. Reaction time was defined as the elapsed time from the end of video playback to the participant’s final response confirmation. Emotion recognition accuracy was computed by comparing each participant’s final response against a ground-truth emotion label pre-assigned to each video clip by the experimenters. Cognitive load was assessed using the Raw NASA-TLX administered after each condition block. We also gathered qualitative feedback about interface preference and the usefulness of the visual cues. All interactions were automatically logged in pseudo-anonymized JSON files for later analysis.

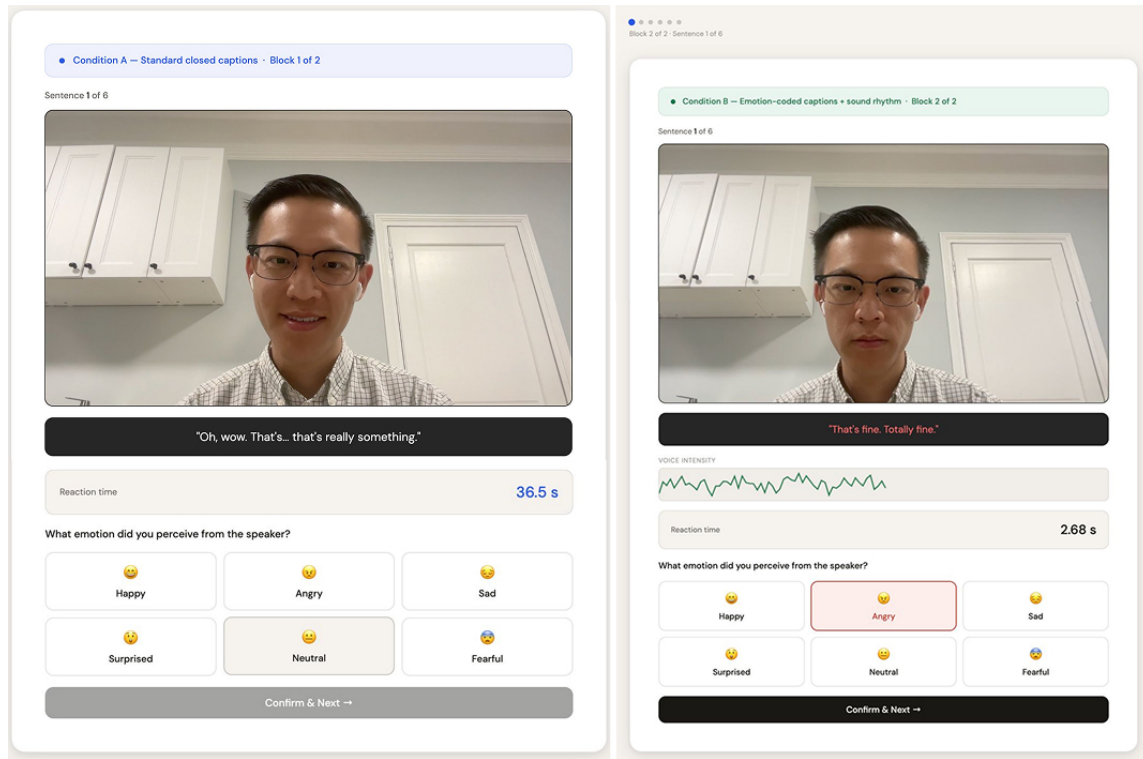


Fig. 2. Comparison of the two experimental interfaces: baseline standard captions and the proposed emotion-coded captions with sound-rhythm visualization.

4.5 Data Collection

All participant actions were recorded through the web platform and exported as pseudo-anonymized JSON logs. The logs included reaction times, selected emotions, final responses, NASA-TLX scores, and debrief answers. This logging approach made it possible to examine both performance-based outcomes and subjective impressions of the interface.

5 Results

Data from 18 participants were included in the analysis. All quantitative comparisons used paired t-tests appropriate to the within-subjects design.

5.1 Reaction Time

Mean reaction time was 3.73s ($SD = 2.11s$) under Condition A and 4.21s ($SD = 3.54s$) under Condition B. The difference was not statistically significant ($t(17) = -0.551, p = .589, d = -0.13$), indicating that the addition of emotion-coded captions and sound-rhythm visualization did not meaningfully affect response speed.

5.2 Emotion Recognition Accuracy

Participants achieved a mean accuracy of 50.9% ($SD = 20.2\%$) under Condition A and 63.0% ($SD = 24.6\%$) under Condition B. The difference was statistically significant ($t(17) = -2.247, p < .05, d = -0.53$), representing a medium effect size. These results suggest that the multimodal interface improved participants' ability to identify the intended emotion of each utterance.

Per-emotion accuracy is shown in Table 1. The largest improvements in Condition B were observed for Angry (+33.3 percentage points) and Sad (+27.8 percentage points), both emotions that are particularly difficult to convey through text alone. Fearful was the most difficult emotion to recognize in both conditions (Condition A: 22.2%; Condition B: 27.8%), while Neutral and Surprised remained high and relatively stable across conditions.

Table 1. Emotion recognition accuracy (%) per condition.

Emotion	Condition A	Condition B	Difference
Happy	66.7	72.2	+5.6
Angry	27.8	61.1	+33.3
Sad	33.3	61.1	+27.8
Surprised	77.8	72.2	-5.6
Neutral	77.8	83.3	+5.6
Fearful	22.2	27.8	+5.6

5.3 Cognitive Load (NASA-TLX)

Mean Raw NASA-TLX total scores were 39.1 ($SD = 15.5$) for Condition A and 44.1 ($SD = 17.5$) for Condition B (scale 0–100, lower = less workload). The difference was not statistically significant ($t(17) = -1.155, p = .264, d = -0.27$), indicating a small and non-significant increase in perceived workload under the multimodal condition.

Table 2 shows mean scores for each NASA-TLX subscale. The largest increases in Condition B were observed for Effort (+12.2 points) and Frustration (+8.6 points), suggesting that the additional visual layer required more deliberate processing and was perceived as less satisfying to use. Temporal demand remained nearly identical across conditions (+0.8 points), indicating that the pacing of the task was not meaningfully affected by the interface.

Table 2. Mean NASA-TLX subscale scores per condition (0–100 scale).

Dimension	Condition A	Condition B	Difference
Mental Demand	45.0	48.3	+3.3
Physical Demand	25.8	28.9	+3.1
Temporal Demand	58.6	59.4	+0.8
Performance	42.2	45.3	+3.1
Effort	31.1	43.3	+12.2
Frustration	30.3	38.9	+8.6

5.4 Qualitative Feedback

Qualitative responses revealed mixed but informative reactions to the two interfaces. Approximately 56% of participants expressed a preference for Condition A, citing lower cognitive demand and fewer competing visual elements. Around

33% preferred Condition B, noting that the emotion colors and sound-rhythm visualization provided useful additional context. The remaining 11% expressed no clear preference.

Emotion color coding was consistently perceived positively across both groups. Participants described the colors as intuitive and reinforcing of caption meaning. One participant noted: *“When the subtitle is red, I think it strengthens the tone, or the person is angry. If it is blue and yellow, I think this sentence is peaceful and relaxed.”*

By contrast, the frequency graph was widely perceived as confusing or distracting. Most participants reported difficulty interpreting it, attributing this to the absence of a legend or scale reference. Several participants stated they ignored it entirely during the task, which aligns with the Effort and Frustration subscale increases observed in the NASA-TLX data.

Participants’ suggested improvements included: adding axis labels or a brief tutorial for the frequency graph (raised by approximately 6 participants), repositioning the emotion selection buttons to reduce scrolling, expanding the emotion taxonomy to cover nuanced states such as sarcasm or disappointment, hiding the countdown timer to reduce time pressure, and increasing the visual salience of emotion colors by using them as caption background rather than text color.

6 Discussion and Conclusion

After analyzing the data, Emotion coded captions produced a statistically significant 12 percent higher emotion recognition accuracy, compared to controlled captions, and the largest effect was on anger (33.3 percent) and sadness (27.8 percent) emotions that are particularly difficult to identify by reading alone. However, the perception of the sound rhythm didn’t add much value: most of the participants had the problem of processing the waveform graph without a legend or scale pointing.

The cognitive load scores (NASA-TLX) were slightly higher with Reso compared to the baseline, indicating the additional visual layer did not decrease the processing load of the waveform graph. This suggests a simple design principle: visual enrichment can only be helpful when the users are able to immediately discern what they are looking at. Also, it is worth mentioning that the higher cognitive load can have other explanations as well. To have a useful evaluation test on the product, we removed the labels from texts since it will immediately give the answer to users, and then the study will not have any value. Furthermore, the existence of a timer in the evaluation will increase the cognitive load and pressure on the users. Consequently, the final product, which includes labels and doesn’t have a timer, may have less cognitive load.

The extent of generalization of these findings has a number of constraints. The sample was all hearing people, who do not have the lived experience of caption fatigue, social exclusion, and coping mechanisms that DHH users develop over time. The one-line video format is also a great reduction of real-life meetings, which involve having multiple speakers, noise in the background, and constant attention demands.

Future directions are to focus on co-design with DHH users, enable users to define emotion-color mappings to accommodate cultural and perceptual variations, and test Reso in live multi-speaker environments. Adding image-based emotion recognition, a more convenient visualization of the rhythm with meaningful labels, and micro-animations depending on the intensity of the voice could be added to the system. On the whole, Reso demonstrates that moving one step further, and bringing tone and affect to sight, is a sound and worthwhile course of action in inclusive communication tools, but the process of figuring out the visual design can take closer collaborations with the communities that they serve.

References

- [1] Chanchal Agrawal and Roshan L Peiris. 2021. I See What You’re Saying: A Literature Review of Eye Tracking Research in Communication of Deaf or Hard of Hearing Users. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, USA) (ASSETS ’21). Association for Computing Machinery, New York, NY, USA, Article 41, 13 pages. doi:10.1145/3441852.3471209
- [2] Robin Angelini, Katta Spiel, and Maartje De Meulder. 2025. Speculating Deaf Tech: Reimagining Technologies Centering Deaf People. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI ’25). Association for Computing Machinery. doi:10.1145/3706598.3713238
- [3] Caluã de Lacerda Pataca, Saad Hassan, Lloyd May, Michelle M Olson, Toni D’aurio, Roshan L Peiris, and Matt Huenerfauth. 2025. Tactile Emotions: Multimodal Affective Captioning with Haptics Improves Narrative Engagement for d/Deaf and Hard-of-Hearing Viewers. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI ’25). Association for Computing Machinery, New York, NY, USA, Article 68, 17 pages. doi:10.1145/3706598.3713304
- [4] Caluã de Lacerda Pataca, Matthew Watkins, Roshan Peiris, Sooyeon Lee, and Matt Huenerfauth. 2023. Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI ’23). Association for Computing Machinery, New York, NY, USA, Article 831, 15 pages. doi:10.1145/3544548.3581511
- [5] Marianna Di Gregorio, Monica Sebillio, Giuliana Vitiello, Antonio Pizza, and Federico Vitale. 2019. ProSign Everywhere - Addressing Communication Empowerment Goals for Deaf People. In *Proceedings of the 5th EAI International Conference on Smart Objects and Technologies for Social Good* (Valencia, Spain) (GoodTechs ’19). Association for Computing Machinery, New York, NY, USA, 207–212. doi:10.1145/3342428.3342695
- [6] Ryo Iijima, Akihisa Shitara, Sayan Sarcar, and Yoichi Ochiai. 2021. Word Cloud for Meeting: A Visualization System for DHH People in Online Meetings. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, USA) (ASSETS ’21). Association for Computing Machinery, New York, NY, USA, Article 99, 4 pages. doi:10.1145/3441852.3476547
- [7] Dhruv Jain, Rachel Franz, Leah Findlater, Jackson Cannon, Raja Kushalnagar, and Jon Froehlich. 2018. Towards Accessible Conversations in a Mobile Context for People who are Deaf and Hard of Hearing. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility* (Galway, Ireland) (ASSETS ’18). Association for Computing Machinery, New York, NY, USA, 81–92. doi:10.1145/3234695.3236362
- [8] Jestin Joy, Kannan Balakrishnan, and Sreeraj Madhavankutty. 2022. SignText: a web-based tool for providing accessible text book contents for Deaf learners. *Universal Access in the Information Society* 21, 3 (Aug. 2022), 717–723.
- [9] JooYeong Kim, SooYeon Ahn, and Jin-Hyuk Hong. 2023. Visible Nuances: A Caption System to Visualize Paralinguistic Speech Cues for Deaf and Hard-of-Hearing Individuals. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI ’23). Association for Computing Machinery, New York, NY, USA, Article 54, 15 pages. doi:10.1145/3544548.3581130
- [10] Yeon Soo Kim, Sunok Lee, and Sangsu Lee. 2022. A Participatory Design Approach to Explore Design Directions for Enhancing Videoconferencing Experience for Non-signing Deaf and Hard of Hearing Users. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility* (Athens, Greece) (ASSETS ’22). Association for Computing Machinery, New York, NY, USA, Article 47, 4 pages. doi:10.1145/3517428.3550375
- [11] Raja S. Kushalnagar and Christian Vogler. 2020. Teleconference Accessibility and Guidelines for Deaf and Hard of Hearing Users. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, Greece) (ASSETS ’20). Association for Computing Machinery, New York, NY, USA, Article 9, 6 pages. doi:10.1145/3373625.3417299
- [12] Emma J McDonnell, Ping Liu, Steven M Goodman, Raja Kushalnagar, Jon E Froehlich, and Leah Findlater. 2021. Social, environmental, and technical: Factors at play in the current use and future design of small-group captioning. *Proc. ACM Hum. Comput. Interact.* 5, CSCW2 (Oct. 2021), 1–25.
- [13] Emma J McDonnell, Soo Hyun Moon, Lucy Jiang, Steven M. Goodman, Raja Kushalnagar, Jon E. Froehlich, and Leah Findlater. 2023. “Easier or Harder, Depending on Who the Hearing Person Is”: Codesigning Videoconferencing Tools for Small Groups with Mixed Hearing Status. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI ’23). Association for Computing Machinery, New York, NY, USA, Article 780, 15 pages. doi:10.1145/3544548.3580809
- [14] Adele Vogel and Jessica L Korte. 2024. What Factors Motivate Culturally Deaf People to Want Assistive Technologies?. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (CHI EA ’24). Association for Computing Machinery. doi:10.1145/3613905.3650871
- [15] Yiwen Wang, Ziming Li, Pratheep Kumar Chelladurai, Wendy Dannels, Tae Oh, and Roshan L Peiris. 2023. Haptic-Captioning: Using Audio-Haptic Interfaces to Enhance Speaker Indication in Real-Time Captions for Deaf and Hard-of-Hearing Viewers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI ’23). Association for Computing Machinery, New York, NY, USA, Article 781, 14 pages. doi:10.1145/3544548.3581076
- [16] Hanaa ZainEldin, Samah A Gamel, Fatma M Talaat, Mansourah Aljohani, Nadiah A Baghdadi, Amer Malki, Mahmoud Badawy, and Mostafa A Elhosseini. 2024. Silent no more: a comprehensive review of artificial intelligence, deep learning, and machine learning in facilitating deaf and mute communication. *Artif. Intell. Rev.* 57, 7 (June 2024).